

“Relative Histories” Formulation of Quantum Mechanics II: Interpretational Issues

W. David Strayhorn, MD, PhD*

Department of Neurology, Washington University School of Medicine, Saint Louis, MO 63110

(Dated: January 6, 2005)

The relative histories formulation (RHF) is a novel formulation of quantum mechanics based upon the assumption that the physical state of the universe can be represented mathematically by an ensemble E of four-dimensional manifolds $\mathcal{W}$. In a companion paper (Part I [26]), the basic formalism of the RHF is presented and demonstrated explicitly to make the same predictions as any of the other standard formulations of quantum mechanics. In the current paper (Part II), we will argue that the RHF is a *single formulation* that lends itself to *two interpretations*: one motivated by positivism, the other by realism. In addition, we use the basic framework of the RHF for an analysis of some of the most notoriously difficult “interpretational” issues in quantum mechanics: locality, causality, completeness, and probability. We find that each of these issues can be addressed in multiple ways – that is, by viewing them in the context of different spaces: $\mathcal{O}$ -State space, $\mathcal{W}_{approx}$, or $\mathcal{W}_{\vec{\gamma}}$. By doing so, the RHF offers multiple perspectives on each of these issues, and demonstrates (for example) how it is that Nature can be simultaneously consistent with the requirements of Einstein *locality* and Bell *nonlocality*. Finally, we compare and contrast the RHF with several other established formulations and interpretations of quantum mechanics, including the consistent histories formulation, Everett’s relative state formulation, Einstein’s ensemble formulation, Feynman’s path integral approach, and Cramer’s Transactional Interpretation.

PACS numbers: PACS numbers go here. Abbreviations: CHF, consistent histories formulation; RHF, relative histories formulation; QM, quantum mechanics; GR, general relativity; FPI, Feynman path integral; MWI, multiple worlds interpretation; EPR, Einstein-Podolsky-Rosen; dBB, de Broglie-Bohm; TIQM, Transactional Interpretation of Quantum Mechanics; CPC, chronology protection conjecture; ER, ensemble-reality interpretation of the RHF; MR, manifold-reality interpretation of the RHF; HUP, Heisenberg Uncertainty Principle

I. INTRODUCTION

The purpose of the present paper is to discuss the interpretational issues surrounding the relative histories formulation (RHF). An introduction to the basics of the RHF is presented in a companion paper (Part I [26]), and is reviewed below. In this paper (Part II), we will argue that the RHF lends itself to a constellation of interpretational advantages that are not found in any other single formulation of quantum mechanics. These advantages may be summarized as follows: (1) a clear definition of the quantum mechanical observer and of its space of states, as well as a clear relationship between the observer and the system being observed; (2) movement of the observer through state space that obeys classical notions of locality and probability; (3) a derivation of quantum statistics and quantum nonlocality from classical notions of probability and locality; (4) a demonstration of compatibility with general relativity; (5) an adherence to the notion that “all is geometry;” (6) the absence of a requirement for any extra “unphysical” dimensions of spacetime beyond the four of our everyday experience; and (7) a spacetime structure that is, at least on the large scale, causal. In addition, we demonstrate that the RHF is consistent with the requirements of Einstein-locality as well as Bell nonlocality, and furthermore, that the RHF provides a novel conceptual tool for understanding the difference between the two. In addition, the RHF provides a novel tool for understanding the EPR argument that quantum mechanics

cannot be considered both complete *and* local (in the sense of Bell-locality). In exchange for these interpretational features, we are forced merely to accept that the structure of spacetime is multiply connected and acausal *on the small scale*.

We will assume that the reader is already familiar with the basic structure of the RHF as presented in [26], in which we demonstrate explicitly that the RHF provides a basis for derivation of quantum statistics. For the purposes of this paper – an analysis of interpretational issues – we will concern ourselves primarily with the structure of the RHF as outlined in Sec. II: Basic Framework of [26], and will not be so concerned with the details regarding the 4-geon model of particles as outlined in Sec. III of that paper. (For reference, a review of the basics of the RHF is presented below. In addition, an overview of the various mathematical entities and spaces of the RHF is depicted in Fig. 1.) One of the features of the RHF is that it allows us to consider some of the most difficult interpretational issues of quantum mechanics – notably, locality and causality – from the viewpoint of different spaces of the RHF. For example, questions of locality may be analyzed separately through an analysis of (1) movement of the observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ through $\mathcal{O}$ -State space; (2) the evolution of the state of an object under observation, like a particle $\mathcal{O}^P$, in $\mathcal{W}_{\vec{\gamma}}$; or (3) the evolution of the state of an object under observation, like a particle $\mathcal{O}^P$, in $\mathcal{W}_{approx}$. As we discuss below, these different types of evolution through these separate spaces are consistent, respectively, with: (1) “RHF locality;” (2) Einstein-locality; and (3) Bell nonlocality. This consistency with Bell nonlocality is the most difficult of these three types of locality to conceptualize. In this paper, we will investigate Bell nonlocality via an analysis of the famous EPR experiment from the framework of the RHF. Once these dif-

*Electronic address: Wstrayho@im.wustl.edu, straycat_md@yahoo.com

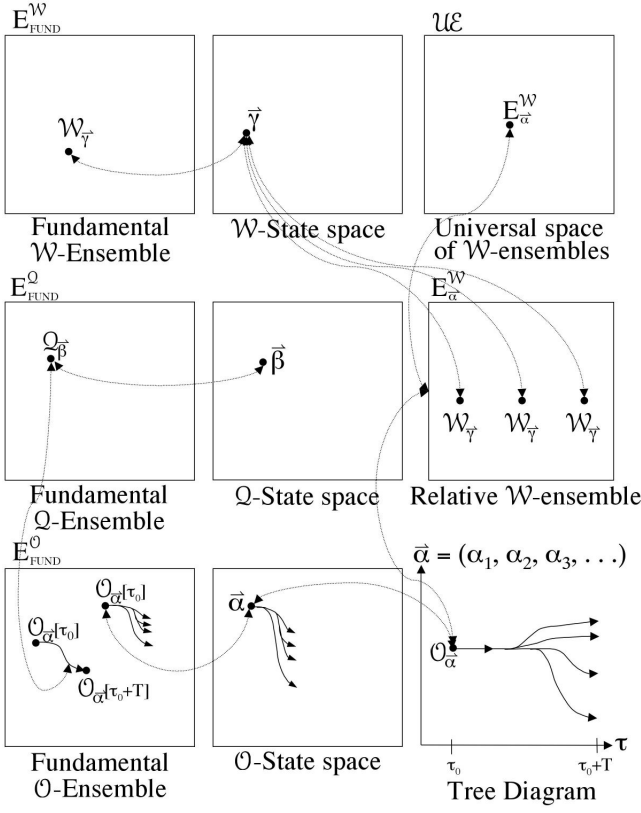


FIG. 1: Overview of the various spaces that are used in the RHF. The fundamental $\mathcal{W}$ -ensemble contains one “copy” of every possible world manifold. A “relative $\mathcal{W}$ -ensemble” is defined in terms of the object manifold $\mathcal{O}_{\vec{\alpha}}$. In this case, any given element $\mathcal{W}$ will typically be represented more than once. In [28], we show that $\mathcal{UE}$, the “universal space of ensembles,” is an infinite-dimensional Hilbert space. See [28] for a further discussion of the properties of these spaces.

ferent types of locality are understood from the framework of the RHF, it is a simple matter to see how they are, in fact, based upon *different definitions* of locality. The standard interpretations do not lend themselves so easily to a differentiation between these different types of locality.

A. Review of the basics

Briefly, the RHF is based upon the assumption that the physical state of any physical object, macroscopic or microscopic, can be represented by a three dimensional manifold $\mathcal{O}$ that is a solution to Einstein’s equations of general relativity. Given any choice of $\mathcal{O}$, there is a unique series of paths through state space called the “tree diagram” that can be constructed according to the manner presented in [26]. The time dependent evolution of the state of this object is represented by the movement of the object through state space, that is, down one of the branches of the tree diagram. At any given branch point, the probability of evolution down any particular

branch is prescribed by the random walk hypothesis, according to which progression down any branch of a branch point with N branches occurs with probability $1/N$. We furthermore assume that *any* physical object may play the role of the quantum mechanical “observer.” Given any designation of $\mathcal{O}_{\vec{\alpha}}$ as the observer, the wavefunction of the Universe is defined as the unique *ensemble* $E_{\vec{\alpha}}^{\mathcal{W}}$ of unbounded four dimensional manifolds $\mathcal{W}_{\vec{\gamma}}$ that satisfy the boundary condition set by $\mathcal{O}_{\vec{\alpha}}$. The evolution of $E_{\vec{\alpha}}^{\mathcal{W}}$ is therefore determined by the evolution of the observer through state space. The states of subsystems of the universe, such as a particle or an experimental apparatus, are likewise represented as three dimensional manifolds $\mathcal{O}^P$ that are submanifolds of the four-manifolds $\mathcal{W}$ that populate the ensemble $E_{\vec{\alpha}}^{\mathcal{W}}$. The state of these subsystems are also defined *in relation* to the observer. For example, if each element of $E_{\vec{\alpha}}^{\mathcal{W}}$ contains a submanifold $\mathcal{O}^P$ that can be identified as a particle at some location in spacetime $a(x, t)$ with respect to the observer, but the exact details of the state of the particle differ among the different elements of $E_{\vec{\alpha}}^{\mathcal{W}}$, then we say that the particle exists in a *superposition* of these various states.

B. “Interpretation” versus “formulation”

Quantum mechanics, like classical mechanics, is a single theory that has multiple mathematical formulations as well as multiple conceptual interpretations [25]. Some notable examples include the Schrödinger wavefunction formulation, Heisenberg’s matrix formulation, the Feynman path integral (FPI) formulation [12] (also known as the “sum over paths”), the consistent histories (or decoherent histories) formulation [20] [21], Everett’s relative state formulation [10] (sometimes referred to as the Many Worlds Interpretation, or MWI [22]), and the de Broglie-Bohm (dBB) formulation [4] [5]. Different *formulations* differ from one another in the sense that they operate using mathematical formalisms that are in some way distinct from one another. Different *interpretations*, on the other hand, differ from one another in the sense that they provide different intuitive “pictures” of the underlying “reality.” This distinction between a “formulation” and an “interpretation” of quantum mechanics is not always clear. This is not surprising, since the mathematics influences the intuitive conceptual picture, and vice versa. It is even possible in some cases (such as the RHF, as we see below) to consider multiple interpretations of a single formulation. The various formulations and interpretations all share the common feature, however, of making the same predictions for any given experiment. Thus, experiment cannot tell us *which* of two interpretations or formulations is the “better” one. From the strict standpoint of experimentation, it is legitimate to adopt whichever interpretation or formulation one would like.

In the development of the RHF, we have adopted what may be described an egalitarian view of the various established interpretations and formulations of quantum mechanics. We realize that not everyone shares this view! Many people have their own favorite interpretation, and find others unpalatable. We find, however, that different interpretations and formulations tend to offer unique insights into quantum mechanics as

a whole, at the mathematical as well as the intuitional levels. In this paper, we will discuss some of the relationships between the RHF and some of these other established formulations. We find that many of the notions of the RHF can be seen to have a counterpart in one or another of the nine or so formulations [25] of QM. Whatever your favorite interpretation or formulation of QM is, the RHF has “borrowed” one or more concepts from it. We will see that a knowledge of these various formulations and interpretations can help in the understanding of the RHF – and perhaps vice versa as well.

C. What’s in a name?

The term “relative” is intended to have several connotations. First, the RHF is formulated using the mathematical language of general relativity, i.e., differential geometry. In particular, the basic elements of the RHF are four-dimensional manifolds $\mathcal{W}$, endowed with metrics g_{ab} and other tensor fields T_{ab} . The RHF is therefore easily made compatible with general relativity, simply by restricting the manifolds $\mathcal{W}$ by Einstein’s equation. Second, the relative histories formulation is similar to Everett’s relative state formulation [10] in the sense that in both cases, the state of a system under observation is formally defined in relation to, or *relative* to, the observer. The term “histories” is meant to refer to the consistent histories formulation [20] [21] (CHF) of quantum mechanics as outlined by Griffiths, Omnès, and others, to which the proposed scheme is closely related. In particular, the proposed scheme may be viewed as a specific example of the consistent histories approach in which a “history” is defined as an ensemble E of manifolds $\mathcal{W}_{\vec{\gamma}}$. (See Table II.)

Our “egalitarian” viewpoint extends not only to the various interpretations of QM, but also to the underlying philosophy of GR. An analogy may be made between a frame of reference in GR and an observer in the RHF. In GR, the requirement of Lorentz invariance may be interpreted as meaning that there is no such thing as a “preferred” frame of reference. Likewise, in the RHF, we adopt the view that there is no such thing as a “preferred” observer. In other words, in the RHF, *any* real, physical object may play the role of the quantum mechanical observer, in much the same way that *any* frame of reference may play the role of the observer’s frame of reference in GR. This is one further meaning of the word “relative” in the title “relative histories formulation.”

D. Organization of the present work

The outline of the present work is as follows. As stated earlier, competing “interpretations” of quantum mechanics differ in the sense that they provide distinct pictures of the “reality” that underlies a theory. To provide an interpretation, therefore, one must provide a clear and explicit *definition* of “reality.” In Sec. II, we will present *two* interpretations of the relative histories formulation, each one based upon a different notion of “reality.” We will alternately use the one (“manifold-reality,” or MR) or the other (“ensemble-reality,” or ER) interpretation

in the discussion that follows. In Sec. III and Sec. IV, we will discuss the issues of locality and completeness from the perspective of each of these interpretations. In particular, we will argue that the MR interpretation provides a realist description of reality that is local and incomplete, whereas the ER description (which is closer than the MR to the standard Copenhagen interpretation) provides a positivist description of reality that is complete but nonlocal. (See Table I.) In this manner, the RHF offers a novel approach for understanding the EPR argument that QM cannot be simultaneously complete and local. In Sec. V, a few miscellaneous issues are discussed: the “speed of collapse” and the meaning of “entanglement” from the perspective of the RHF. Finally, in Sec. VI, we compare the RHF to several other formulations of QM: the consistent histories formulation, Everett’s relative state formulation (the MWI), Einstein’s ensemble formulation, and the FPI.

II. TWO DEFINITIONS OF “REALITY” WITHIN THE RELATIVE HISTORIES FORMULATION

We will present two interpretations – the “manifold reality” interpretation and the “ensemble reality” interpretation – and will use the famous experiment of Schrödinger’s cat to illustrate how they differ. In each of these, “reality” is defined relative to an observer – that is, relative to the 3-manifold $\mathcal{O}_{\vec{\alpha}}$ that is used to define the ensemble $E_{\vec{\alpha}}^{\mathcal{W}}$. For the Schrödinger’s cat experiment, $E_{\vec{\alpha}}^{\mathcal{W}}$ would be a union of disjoint subsets $E_d^{\mathcal{W}}$ and $E_a^{\mathcal{W}}$ that consist of worlds in which the cat is either dead or alive, respectively.

Definition 1. “*Ensemble-Reality*” Interpretation. *With respect to an observer $\mathcal{O}_{\vec{\alpha}}$, “reality” is defined as the entire ensemble, $E_{\vec{\alpha}}^{\mathcal{W}}$. (See Fig. 1.)*

In other words, every manifold $\mathcal{W}$ that is an element of the ensemble $E_{\vec{\alpha}}^{\mathcal{W}}$ is considered to be an *equally valid* description of “reality.” Thus, if one asks whether statement X is true “in reality,” then the answer is determined by evaluating statement X with respect to $\mathcal{O}_{\vec{\alpha}}$ (as outlined in Part I [26]). Therefore, we would say – according to this interpretation – that in reality, the cat is in a genuine superposition of alive and dead states, where we mean superposition in the usual (non classical) sense of the word.

We may make use here of the four-manifold $\mathcal{W}_{approx}$ to clarify what we mean by “superposition.” Recall that given any choice of the observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$, there is an associated *ensemble* of manifolds $\mathcal{W}_{\vec{\alpha}}$, but just *one* $\mathcal{W}_{approx}$. The purpose of $\mathcal{W}_{approx}$ is to serve as a sort of “background spacetime” upon which to map the various objects under observation, such as a cat or a particle, from each of the $\mathcal{W}_{\vec{\alpha}}$ ’s in $E_{\vec{\alpha}}^{\mathcal{W}}$. Some $\mathcal{W}_{\vec{\alpha}}$ ’s in $E_{\vec{\alpha}}^{\mathcal{W}}$, for example, would map a dead cat to some spacetime location in $\mathcal{W}_{approx}$; others would map a live cat – *to the same location* – in $\mathcal{W}_{approx}$. The cat is in a superposition by virtue of the fact that it is in a different state (dead, alive) in different $\mathcal{W}_{\vec{\alpha}}$ ’s, even though in each case it maps to the same location in $\mathcal{W}_{approx}$. In this manner, $\mathcal{W}_{approx}$ is populated by a myriad of objects (cats, particles) that exist in a “superposition” of various different states. Of course, the

“state” of some object in question may correspond to the position operator, in which case different $\mathcal{W}_\alpha$ ’s would result in a mapping of the object in question to *different locations* in $\mathcal{W}_{approx}$.

What role does $\mathcal{W}_{approx}$ play in the ER interpretation of the RHF? For a crude visualization, using the cat as an example, we may imagine that in $\mathcal{W}_{approx}$, there are two cats, one alive and one dead, existing at the same location $a(x, t)$ in ghostlike superposition. In the framework of the RHF, at the instant that the observer encounters the bifurcation in the tree diagram into the “cat alive” branch and the “cat dead” branch, the cat will instantly change from this ghostlike superposition into either the alive or dead state. (Although see Sec. V A for a discussion of whether or not collapse is instantaneous.) This situation is not unlike that offered by the standard Copenhagen interpretation. The ER, therefore, is similar in spirit to the Copenhagen interpretation; we say that the “real state” of the cat is represented by the *complete set* of each the individual states represented by each of the cats in each of the $\mathcal{W}_\alpha$ ’s in $E_\alpha^\mathcal{W}$. The instant of collapse corresponds to the depopulation of this set from multiple elements (alive and dead) to, in this case, just one element (dead or alive).

Definition 2. “Manifold-Reality” Interpretation. Assume an observer $\mathcal{O}_\alpha$ and its corresponding ensemble $E_\alpha^\mathcal{W}$. Within $E_\alpha^\mathcal{W}$, there exists a single, unique, “special” manifold $\mathcal{W}_{real}$ that corresponds to the “real” description of the universe. (See Fig. 1.)

According to this view, we would say that the cat is *either* alive *or* dead, where we mean “either” in the *classical* sense. Whether the cat is truly alive or dead depends on *which* manifold $\mathcal{W}$ is the *real* manifold. However, there is no method (at $\tau = 0$, at least) by which the observer $\mathcal{O}_\alpha$ is able to identify this one special manifold out of the ensemble $E_\alpha^\mathcal{W}$. That is, the identity of the true manifold remains hidden from the observer. Therefore, as long as $E_\alpha^\mathcal{W}$ exists as a union of nonempty sets $E_a^\mathcal{W}$ and $E_d^\mathcal{W}$, we will be *unable in principle* to determine whether the cat is alive or dead.

These interpretations are interesting from a “philosophical” perspective. However, we emphasize that the choice of definition of “reality” that one adopts in this context will have no bearing upon the mathematical formalism itself, or on the *experimental predictions* that are made by the formulation.

A. Philosophical motivations underlying the MR and ER

We note that the ER interpretation is essentially motivated by a *positivist* philosophy, according to which there is, in principle (or by definition), no element of reality that is *inaccessible to observation*. As such, the standard Copenhagen interpretation – as we have already seen – is probably better represented by the ER interpretation than by the MR interpretation. The MR interpretation, on the other hand, is essentially motivated by a *realist* philosophy, according to which there *can* exist elements of reality that are inaccessible to an observer. Indeed, the MR interpretation is compatible with the notion that any given particle does in fact have a definite position

TABLE I: Comparison between the two *interpretations* of the relative histories formulation.

	Manifold Reality (MR)	Ensemble Reality (ER)
Definition of reality relative to $\mathcal{O}_\alpha$	Reality is one special manifold, $\mathcal{W}_{\vec{\gamma}}$ which is an element of $E_\alpha^\mathcal{W}$.	Reality is the entire ensemble $E_\alpha^\mathcal{W}$.
Philosophical motivation	Realism	Positivism
Is a complete description of Reality accessible to the observer?	No.	Yes.
Does the RHF provide a local description of Reality?	Yes (Einstein locality)	No (Bell nonlocality)
What is the hidden variable?	$\vec{\gamma}$	There isn’t one.

and definite momentum at any given point in time, and that the uncertainty in these variables is merely a reflection of our *ignorance*.

III. LOCALITY AND NONLOCALITY

One of the most difficult interpretational issues of modern physics is the issue of locality. Since the advent of quantum mechanics, and notably since the important contributions of Bell [3], it has been generally understood that QM is a non-local theory. Nevertheless, general relativity is a purely local theory of nature. Although this appears to present a contradiction between QM and GR, it is not clear that there is in fact any explicit contradiction. In our view, there is in fact no contradiction. Rather, there are actually at least two different types of locality: Bell-locality and Einstein locality. Part of the difficulty is that it is difficult to conceptualize the difference between these two types of locality.

In this section, we discuss how to interpret the phenomena of locality and nonlocality from the point of view of the RHF. We will see that the RHF is capable of reproducing, as it must, Einstein locality as well as EPR nonlocality. We will also see that the RHF presents a conceptual tool to differentiate between these different types of locality; namely, each different type of locality (or nonlocality) is manifest within a different space of the RHF.

A. EPR-Nonlocality within the RHF: The EPR experiment

Any formulation of quantum mechanics must be capable of reproducing the results of the famous EPR experiment. Fig 2 outlines a proposed representation of the EPR experiment from the framework of the RHF. Let us make clear that this proposed analysis will make use of the basic structure of

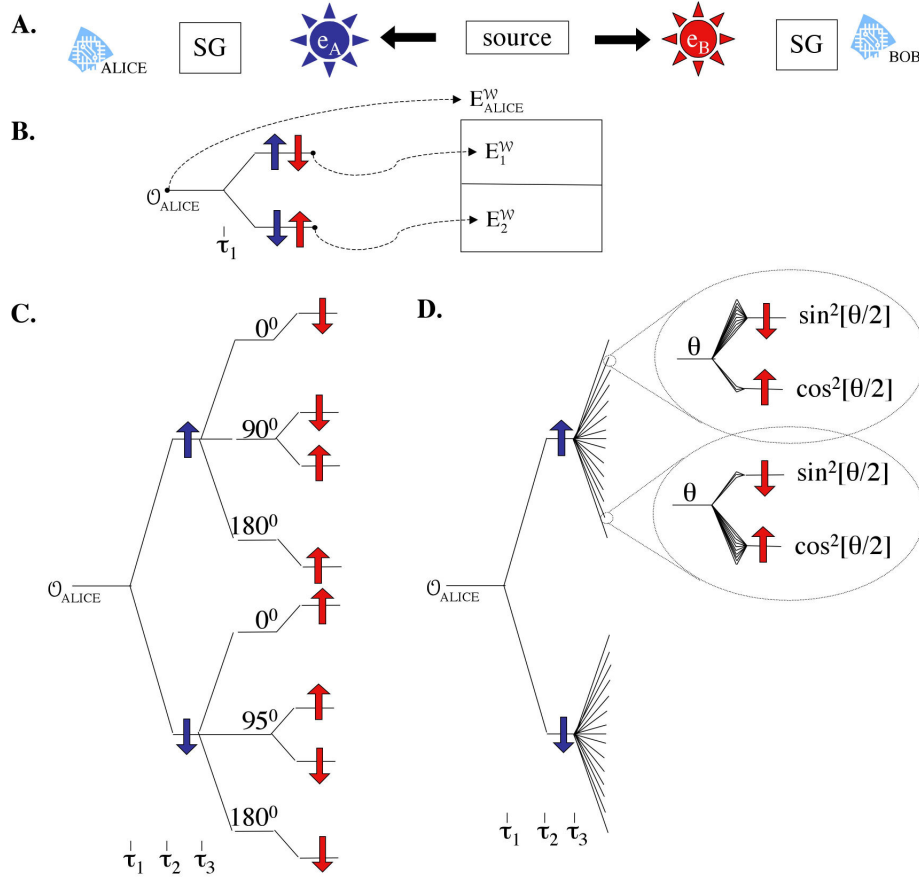


FIG. 2: **A hypothetical relative histories representation of the EPR experiment.** **A.** Experimental setup. (SG: Stern-Gerlach apparatus.) Alice maintains the angle of her SG apparatus constant, whereas Bob varies the angle of his SG apparatus randomly at the last possible instant to an angle θ relative to Alice's apparatus. **B.** Bob maintains $\theta = 0$. **C.** Bob varies θ randomly between 0, 90, and 180 degrees. **D.** Same as in C except that Bob varies θ among a large number of possibilities. Blow-ups: magnified views of two separate regions of the tree diagram.

the RHF (Sec. II of [26]), but will *not* make use of the topological properties of 4-geons presented in Sec. III of that paper. Indeed, we have not, as of yet, presented a general model of how the RHF should model spin. (We make note, though, that Hadley has in fact presented a 4-geon model [18] of spin half particles that is consistent with the general framework of the RHF. It will be interesting to investigate what implications his model may have on the workings of the RHF.) The purpose of this section is not to demonstrate that the RHF reproduces the *exact* statistics of the EPR experiment, but rather merely to demonstrate that the RHF is capable of reproducing the *nonlocality* that is inherent in the EPR experiment. The issue of the reproduction of quantum statistics in all of its glorious detail by the RHF is addressed in Part I [26].

The basic experimental setup of a typical EPR experiment is outlined in Fig 2 A. In this experiment, there is a source that emits a correlated pair of electrons, e_A and e_B , that travel toward observers Alice and Bob, respectively. Alice and Bob each have a Stern-Gerlach (SG) apparatus that measures the spin state of the electron along a specific axis. For simplicity

and without loss of generality, we assume that Alice's SG apparatus is fixed at a given angle, and the angle of Bob's SG apparatus can be set to any angle θ with respect to Alice's apparatus.

In Fig 2 B, Bob has fixed θ to be zero degrees. On the left is depicted the tree diagram of the experiment using Alice as the "observer." We use the 3-object $\mathcal{O}_{ALICE}$ to represent the state of Alice's computer at the instant of emission of the electron pair by the source. At a specific time τ_1 , Alice measures the spin state of her electron. This measurement event corresponds to the bifurcation in the tree diagram of Fig 2 B. One branch corresponds to the measurement by Alice that her electron is spin up; the other, spin down. Since these are mutually exclusive events that cannot coexist within the same manifold $\mathcal{W}$, each of these branches will map to disjoint ensembles $E_1^{\mathcal{W}}$ and $E_2^{\mathcal{W}}$. Since there are no instances in which both electrons are measured to be up or both measured to be down, we know that for every $\mathcal{W} \in E_1^{\mathcal{W}}$, Bob will measure his electron to be spin down; for every $\mathcal{W} \in E_2^{\mathcal{W}}$, Bob will measure his electron to be spin up. (Actually, it may be better

to say that there *do* exist manifolds in which both electrons are measured to be spin up or both are measured to be spin down, but that the probability measure associated with collapse onto one of these possibilities is vanishingly small. This appears to be the case when one considers the situation depicted in Fig 2 D, discussed below, and takes the limiting case that $\theta = 0$.)

Let us consider this from the perspective of the ensemble reality (ER) interpretation. Immediately prior to Alice’s measurement, and then immediately after her measurement, what is the “real state” of Bob’s electron? Immediately prior, e_B is in a superposition; immediately after, it is in one of the definite spin states. Thus, the *real state* of e_B changes precisely at the instant of Alice’s measurement of e_A . (Recall also our discussion of the role of $\mathcal{W}_{approx}$ in the ER interpretation. At some location in $\mathcal{W}_{approx}$, the state of Bob’s electron is represented by the set: (up, down). At the time of Alice’s measurement, this set changes to have only one element, either (up) or (down).)

Consider now a modification of the above experiment such that immediately prior to τ_1 , Alice picks among two options: either measure the spin state of e_A , or forego the measurement. If she makes the measurement, she will induce an *immediate* change in the state of e_B from a superpositional to a definite spin state. If she foregoes the measurement, e_B will remain in the superpositional state. Thus we see that the state of e_B at position $x = x_B$ and time τ_1 depends upon an event at position $x = x_A$, also at time τ_1 . (Time and distance intervals measured from Alice’s frame of reference.) Thus, we have an event at spacetime location (x_A, τ_1) (Alice’s decision whether or not to measure the spin state of her particle) that influences the state of a particle at spacetime location (x_B, τ_1) . (x_A, τ_1) and (x_B, τ_1) are spacelike separated locations. Therefore, when viewed from the ER perspective, the RHF supports nonlocal interactions.

From the perspective of the MR interpretation, in the above scenario, the real state of Bob’s particle is definite spin-up or spin-down at spacetime location (x_B, τ_1) regardless of whether or not Alice measures the spin state of her particle. The state of Bob’s particle, of course, depends upon *which* individual $\mathcal{W}_{\vec{\gamma}}$ in $E_{\vec{\alpha}}^{\mathcal{W}}$ is the “real” manifold, $\mathcal{W}_{real}$. And the identity of $\mathcal{W}_{real}$ is forever fixed; it does not change as a result of measurement. But of course, we *don’t know* which $\mathcal{W}_{\vec{\gamma}}$ is $\mathcal{W}_{real}$, and in principle, we cannot. The point is that in the MR interpretation, Bob’s particle does not undergo any change in its state as a result of measurement. Therefore, within the confines of the MR interpretation, the above scenario does not provide an example of nonlocal interactions.

B. Aspect modification of EPR

The EPR experiment may be modified by allowing the polarization angle of the SG apparatus to vary. In Fig 2 C, the experiment is set up so that Alice’s SG apparatus is held at a constant angle, and Bob is allowed to vary the angle of his SG apparatus to an angle θ 0, 90, or 180 degrees relative to Alice’s apparatus. Consistency with Hawking’s chronology protection conjecture (see Part I [26]) demands that the con-

figuration of Alice’s tree diagram at any given point in her proper time τ can be a function of θ *only* if the value of θ can be communicated, in the classical sense, to Alice. We assume that Bob chooses this angle randomly and does so at the last possible instant so that there is no possibility that the state of his apparatus – i.e., the angle θ – can be communicated (in the classical sense) to Alice by the time of her measurement. Let us imagine, therefore, that the setting of Bob’s apparatus (θ) as well as the results of Bob’s measurement are communicated to Alice at the speed of light (via, for instance, a powerful telescope that Alice uses to spy on Bob’s laboratory.) Since Bob’s SG apparatus *must* be set to the angle θ immediately *prior* to his measurement, Alice will observe θ immediately *prior* to her observation of Bob’s spin measurement. In other words, consistency with the CPC demands that Alice’s tree diagram results in collapse onto θ *prior* to collapse onto a specific spin state for e_B . This is reflected in her tree diagram, in which we have collapse in the following order: first, collapse (at τ_1) onto a specific spin state for e_A ; second, collapse (at τ_2) onto a specific value for θ ; and third, collapse (at τ_3) onto a specific spin state for e_B .

We see that the tree diagram predicts the same probabilities as standard quantum mechanics. For example: Alice predicts a 50 percent chance that she will observe her electron to be spin up or spin down. We assume that Bob chooses θ randomly; this is reflected in the tree diagram which predicts a 33 percent chance of collapse onto, say, $\theta = 90$ degrees. In the case that Alice observes e_A to be spin up and θ to be 90 degrees, the tree diagram predicts 50 percent chance that Bob will communicate to her that his electron is spin up.

We can further generalize this experiment by allowing Bob to vary θ among an arbitrarily large number of possible angles, as depicted in Fig 2 D. This figure depicts “blow-up” magnifications of two areas of the tree diagram. In these areas, the trajectories of the tree diagram involve a splitting followed by a fusion into either a spin-up state or a spin-down state. Following the dictates of the random walk hypothesis, the relative probability of collapse onto either one of these two spin states is proportional to the number of trajectories at the splitting. Since Alice has already “observed” the value of θ by τ_2 , then it is plausible that the branch structure of the tree diagram at τ_3 can be a function of θ without violation of Hawking’s chronology protection conjecture (CPC; see Part I [26]). We have depicted these functions to the right of the figure and have chosen them to reflect the predictions of standard quantum mechanics. Note that in this analysis, we have implicitly *assumed* that the tree diagram will give rise to quantum statistics. We emphasize that our goal in this section is not to demonstrate that the RHF *must* give rise to quantum statistics, but merely to demonstrate that it *can*.

In summary, we have demonstrated how it is that the RHF is able to recapitulate the predictions, not only of the original EPR experiment, but also of the delayed-choice modification as originally proposed by ? and later tested by Aspect. We note that the thought process outlined above could easily be extended to the double-delayed choice EPR experiment. Since these experiments are used to demonstrate the Bell-type non-locality of standard quantum mechanics, we see that the RHF

is likewise capable of modelling nonlocality.

As a corollary to this conclusion, consider the following question. From a purely “philosophical” perspective, which is more objectionable: the basic framework of the RHF as outlined in Sec. II of Part I [26], or standard quantum mechanics? If the basic framework of the RHF is deemed “nonobjectionable” from a “philosophical” perspective, then *quantum non-locality must follow suit in being equally nonobjectionable*.

C. Einstein-Locality within the RHF

Any theory of quantum gravity must obey Einstein locality. We will argue that the RHF meets this criterion. Basically, this is because the essential elements of the RHF are 4-manifolds, $\mathcal{W}_{\vec{\gamma}}$. Taken on an individual basis, any given 4-manifold $\mathcal{W}_{\vec{\gamma}}$ is local in the sense that general relativity is a local theory. This could be expressed mathematically as follows. At any given point (x_1, x_2, x_3, x_4) , it is possible to calculate the values of any tensor fields (including the metric) at (x_1, x_2, x_3, x_4) given only the values of these fields at points immediately surrounding (x_1, x_2, x_3, x_4) , that is, within a ball of arbitrarily small radius centered on (x_1, x_2, x_3, x_4) (with the center point removed). Therefore, we say that the RHF obeys “Einstein locality.” We might also say that the RHF is local with respect to the four spatial variables (x_1, x_2, x_3, x_4) .

If we adopt the MR interpretation of the RHF, according to which “reality” corresponds to one particular $\mathcal{W}_{\vec{\gamma}}$, then we say that “reality” is local. In other words, the MR interpretation of the RHF may be considered a local theory of Nature (Table I).

D. RHF-Locality of the movement of an observer through $\mathcal{O}$ -State Space

Let us also consider the flow of an observer $\mathcal{O}_o = \mathcal{O}_{\vec{\alpha}}$ through $E_{FUND}^{\mathcal{O}}$ (or, alternatively, through $\mathcal{O}$ -State Space; or, equivalently along a tree diagram; see Fig. 1). The local structure of the tree diagram – by which we mean the structure of the tree diagram within a ball in state space of arbitrarily small radius, centered about the point $\vec{\alpha}$ – can be calculated completely using $\mathcal{O}_{\vec{\alpha}}$ (or simply $\vec{\alpha}$) as the sole input. Therefore, everything we need to know about the time dependent evolution of $\mathcal{O}_{\vec{\alpha}}$ – including the probabilities of evolving into this or that state – can be calculated using only $\mathcal{O}_{\vec{\alpha}}$ as input. In particular, the evolution of $\mathcal{O}_{\vec{\alpha}}$ is dependent solely on the *local characteristics of $\mathcal{O}$ -state space*, and *independent* of the characteristics of $\mathcal{O}$ -state space at any location other than at $\vec{\alpha}$. In this sense, we say that the RHF is *local with respect to the parameter $\vec{\alpha}$ of $\mathcal{O}$ -state space*. We call this “RHF-locality” to distinguish it from Einstein-locality (which describes $\mathcal{W}_{\vec{\gamma}}$) or Bell-nonlocality (which describes $\mathcal{W}_{approx}$).

E. Summary of locality issues

How is it that the RHF can be local and nonlocal at the same time? Indeed, the same question may be asked of con-

temporary physics! With respect to the RHF, we see that the question: is it local? depends upon the particular mathematical entity with which we are concerned. With respect to any given particular world manifold $\mathcal{W}_{\vec{\gamma}}$, the RHF obeys Einstein-locality. With respect to the movement of the observer through $\mathcal{O}$ -state space, the RHF is also a local theory. However, when one thinks in terms of the ensemble $E_{\vec{\alpha}}^{\mathcal{W}}$, and of the fact that each element of $E_{\vec{\alpha}}^{\mathcal{W}}$ may map a distinct state of the same object onto $\mathcal{W}_{approx}$, as we saw in our discussion of the ER interpretation of the RHF, we see that the RHF obeys EPR-nonlocality. The advantage of the RHF over conventional formulations of quantum mechanics is that it becomes conceptually easier to differentiate these different issues of locality, that is, by categorizing them with respect to different mathematical entities.

IV. COMPLETE OR INCOMPLETE?

Does quantum mechanics provide a *complete* description of reality? This is a question that has been raised since the inception of quantum theory. We note that it is difficult if not impossible even to *ask* this question without a notion of what elements of a theory correspond to “reality.” It is therefore natural that this question might have one answer from the framework of the ER interpretation of the RHF, and a different answer from the framework of the MR interpretation.

Indeed, we find that such is the case. Let us further refine the question of completeness in the following manner: Does the formalism of the RHF provide an observer $\mathcal{O}_{\vec{\alpha}}$ with a method to solve for all variables that correspond to reality? In more slang terminology, we may ask whether the formalism of the RHF allows the observer to “know everything there is to know” about reality? From the standpoint of the MR interpretation, the answer is clearly: no. In the MR interpretation, every particle has a definite position and a definite momentum at all points in time. However, most of this information is *inaccessible* to the observer. The reason for this is that the observer is unable, in principle, to know *which* element of $E_{\vec{\alpha}}^{\mathcal{W}}$ is the “real” one, $\mathcal{W}_{real}$. The observer’s knowledge of reality is, therefore, *incomplete*.

From the standpoint of the ER interpretation, however, the answer is “yes”. This is because it is possible to arrive at a single definite solution to $E_{\vec{\alpha}}^{\mathcal{W}}$, given merely information that is locally accessible to $\mathcal{O}_{\vec{\alpha}}$. And the solution to $E_{\vec{\alpha}}^{\mathcal{W}}$ tells us “all there is to know” about reality. Thus, the observer’s knowledge of reality is *complete*.

A. The choice between completion and locality

It has been argued that Bell’s analysis of the EPR experiment demonstrates that quantum mechanics provides a description of nature that cannot be considered both local *and* complete. In other words, we are forced to give up either locality or completeness, if not both. Table I summarizes the perspectives on these issues offered by the MR and the ER interpretations of the RHF. We see that the RHF underscores the

EPR argument: according to the MR interpretation, nature is local and incomplete; and according to the ER interpretation, it is nonlocal and complete.

V. MISC. FURTHER ISSUES

A. The “speed of collapse”

The process of “collapse” of a variable X is modelled in the RHF as a transition along a tree diagram from a point $\mathcal{O}(\tau_1)$ to a point $\mathcal{O}(\tau_2)$, with $\tau_2 > \tau_1$, such that the evaluation of statement X (for example: “a particle is spin up”) at $\mathcal{O}(\tau_1)$ is in a superposition, whereas the evaluation of statement X at $\mathcal{O}(\tau_2)$ is in a definite state. In the RHF, the evaluation of any (two-valued) statement X at any given point $\mathcal{O}$ *must* be either S, T, or F; there are no other options. Therefore, the instant of “collapse” within the framework of the RHF would be most appropriately defined as the point in proper time τ_c such that for $\tau < \tau_c$, the evaluation of statement X at $\mathcal{O}(\tau)$ is S, whereas for $\tau > \tau_c$, the evaluation of statement X at $\mathcal{O}(\tau)$ is either T or F. Therefore, the “speed of collapse” is considered to be “instantaneous” from the framework of the RHF. From the framework of conventional quantum mechanics, the speed of collapse has been an area of investigation. Several experimenters have attempted to measure a lower bound on the time interval within which collapse occurs. One notable group [24] sets a lower bound of $1.5 \times 10^4 c$; to our knowledge, no *upper* bound has been set. Indeed, the discovery of the existence of any finite upper bound (i.e., less than ∞) would necessitate the introduction of a preferred frame of reference and would therefore constitute an *explicit conflict* with general relativity. This is discussed in (further papers by the Gisin group?), and we will not recapitulate the full argument here. In the current absence of any experiment that has fully settled this question, we will merely note that the notion that collapse is instantaneous – which is the view most consistent with the RHF – is also, at the very least, *compatible* with contemporary physical theory.

B. Entanglement

Two objects are said to be *entangled* from the perspective of an observer $\mathcal{O}_o$ iff there are certain subsets $E^{\mathcal{W}} \in E_o^{\mathcal{W}}$ whose probability measure is either zero or is vanishingly small. For example, consider two electrons e_1 and e_2 whose spin states are anticorrelated. This may be expressed in the terminology of the RHF in the following manner. Define $E_{uu}^{\mathcal{W}}$, $E_{ud}^{\mathcal{W}}$, $E_{du}^{\mathcal{W}}$, and $E_{dd}^{\mathcal{W}}$ as the ensembles of manifolds in which the two particles are in the down, down states (etc). Perfect anticorrelation of the two particles implies that the probability measures of $E_{dd}^{\mathcal{W}}$ and $E_{uu}^{\mathcal{W}}$ are either exactly zero, or are vanishingly small. (Typically, we would define the probability measures of $E_{ud}^{\mathcal{W}}$ and $E_{du}^{\mathcal{W}}$ to be 0.5 and 0.5, respectively.)

VI. COMPARISON OF THE RELATIVE HISTORIES FORMULATION WITH SEVERAL OTHER FORMULATIONS

The basic purpose of making comparisons between the RHF and any other formulation is to make use of these other formulations to provide insight into the further development of the RHF. For example, the requirement that the decoherence functional of the consistent histories formulation (CHF) must be Hermetian [20] [21] may be utilized within the context of the RHF to guide us towards an appropriate definition for the metric over the Hilbert space $\mathcal{UE}$. (See Fig. 1 and Part III [28].) Our essential strategy here is to take a concept or relationship from one of these other formulations and to “translate” it into the terminology of the RHF. Central to our strategy is the assumption that none of the formulations are any more “correct” than any of the other formulations. Each one of them, in our estimation, approaches QM from a different angle and will likely provide its own unique set of insights.

A. Relationship to the Consistent Histories Formulation

The consistent histories formulation (CHF) is a generalized quantum scheme put forward by Griffiths and by Omnés, and developed by others including Gell-Mann, Hartle, and Isham [20] [21]. It is difficult to summarize just what exactly this scheme is, especially considering that there is more than one version of it. An exhaustive comparison between the CHF and the RHF, furthermore, would be beyond the scope of the present work. Therefore, for the reader who is already familiar with the consistent histories formulation, a listing is presented in table format (see Table II) that depicts a proposed correspondence between some of the mathematical entities of the CHF and the RHF. (See Part III [28] for a discussion of the concepts of temporal support $\vec{t}$ and the decoherence functional $d[E_{\vec{\alpha}'}^{\mathcal{W}}, E_{\vec{\alpha}''}^{\mathcal{W}}]$.)

One of the central concepts of the CHF is the notion itself of a history. As far as I can tell from the consistent histories literature, the details regarding the precise mathematical specification of a “history” is still to an extent an open question, and has not been spelled out in as much detail as is presented here for the relative histories formulation. Typically, a history is simply defined as a series of events, for example: a particle is at location a_1 , then a_2 , then a_3 , and so on. However, consideration has been given to the notion that a manifold $\mathcal{M} = (M, g)$ may be considered to be a “history.” I have been so far unable to find this idea developed to the extent or in the manner suggested by the present work. Indeed, it has been argued that this cannot be done. As stated in Ref. ??, “a Lorentzian manifold (M, g) that is not globally hyperbolic cannot be regarded as a time-ordered sequence of three-geometries as there is no global time function with which to construct an appropriate one-parameter family of foliations of $\mathcal{M}$. In this sense, the Lorentzian metric g is not a ‘history’ ...” However, the comparison between the CHF and the RHF presented in Table II is based upon the notion that the RHF is, in fact, a *specific case of the CHF*, in which a

TABLE II: Comparison of some of the notation and concepts of the consistent histories formulation and the relative histories formulation. The notation of the left hand column is taken from Ref. [21]. This table points out similarities (not the differences) between these two formulations.

consistent histories	relative histories
a history	$\mathcal{W}_{\vec{\gamma}}$
α (a “history proposition”)	$E_{\vec{\alpha}}^{\mathcal{W}}$
$\mathcal{UP}$ =set of all history propositions	$\mathcal{UE}$ =set of all $E_{\vec{\alpha}}^{\mathcal{W}}$
$\alpha \leq \beta$	$E_{\vec{\alpha}'}^{\mathcal{W}} \subset E_{\vec{\alpha}''}^{\mathcal{W}}$
decoherence functional $d(\alpha, \beta) : \mathcal{UP} \times \mathcal{UP} \rightarrow \mathbb{C}$	$d[E_{\vec{\alpha}'}^{\mathcal{W}}, E_{\vec{\alpha}''}^{\mathcal{W}}]$
temporal support	$\vec{t}$
Hermiticity: for all α, β , $d(\alpha, \beta) = d(\beta, \alpha)^*$	Hermiticity: for all $E_{\vec{\alpha}'}^{\mathcal{W}}$, $E_{\vec{\alpha}''}^{\mathcal{W}}$, $d[E_{\vec{\alpha}'}^{\mathcal{W}}, E_{\vec{\alpha}''}^{\mathcal{W}}] =$ $d[E_{\vec{\alpha}''}^{\mathcal{W}}, E_{\vec{\alpha}'}^{\mathcal{W}}]^*$

history *is* in fact a manifold, $\mathcal{W}_{\vec{\gamma}}$. In response to the above argument, we make note of the fact that the RHF *does* provide a mechanism for constructing a time-ordered sequence of *four-*geometries $\mathcal{W}_{\vec{\gamma}}$, using the proper time τ as the one-parameter that orders our 4-manifolds.

The consistent histories formulation shares at least one major interpretational motivation with the RHF, that is: “a desire to replace the traditional Copenhagen interpretation of quantum theory with one that avoids any fundamental split between observer and system” [20]. Clearly the RHF avoids any such fundamental split.

One major difference between the CHF and the RHF is that the CHF *assumes* quantum statistics, whereas the RHF *assumes* classical statistics from which we *derive* quantum statistics. (At least, I think so . . . it’s actually not entirely clear to me at what point the concept of probability enters the picture in the CHF . . .)

B. Relationship to Everett’s relative state formulation

My purpose in this section is to illustrate the compatibility between the RHF and Everett’s relative state formulation (i.e., the MWI) by pointing out some of the similarities and correspondences in the concepts of these two formulations. The comparison is by no means exhaustive.

Everett’s concept of the *relativity of states* was summed up in his seminal paper [10] as follows:

“A constituent subsystem cannot be said to be in any single well-defined state, independently of the remainder of the composite system. To any arbitrarily chosen state for one subsystem there will correspond a unique *relative state* for the remainder of the composite system. This relative state will usually depend upon the choice of state for the first subsystem. Thus the state of one subsystem does not have an independent existence, but is fixed only by the state of the remaining subsystem.”

The above quotation makes perfect sense from the standpoint of the RHF if one interprets the “subsystem” as the 3-object, $\mathcal{O}_{\vec{\alpha}}$, that corresponds to our observer, and the “composite system” as its corresponding ensemble, $E_{\vec{\alpha}}^{\mathcal{W}}$. Translation (see Fig. 1):

“Any $\mathcal{O}_{\vec{\alpha}}$ cannot be said to be in any single well-defined state, independently of $E_{\vec{\alpha}}^{\mathcal{W}}$. To any arbitrarily chosen $\mathcal{O}_{\vec{\alpha}}$, there will correspond a unique $E_{\vec{\alpha}}^{\mathcal{W}}$. $E_{\vec{\alpha}}^{\mathcal{W}}$ will usually depend upon the choice of $\mathcal{O}_{\vec{\alpha}}$. Thus the state of $\mathcal{O}_{\vec{\alpha}}$ does not have an independent existence, but is fixed only by $E_{\vec{\alpha}}^{\mathcal{W}}$.”

Once we do the translation, we see that Everett was somewhat repetitive in his statement above.

The multiple worlds interpretation, or MWI, is typically considered to be identical to Everett’s original relative state formulation, although the terminology has evolved somewhat. We refer the reader to [22] for an introduction. I would point out in passing that the diagram intended to illustrate worlds splitting in <http://www.hedweb.com/manworld.htm#differentiate> (section 19) is essentially an example of a tree diagram in the RHF. In section 17, <http://www.hedweb.com/manworld.htm#fuse> Price addresses the issue of “fusion” verses “splitting.” The view of the MWI (according to Price) is that worlds *can* fuse, although considerations of thermodynamics lead to the conclusion that “in the forward time direction” they “*almost* never fuse” (*italics mine*). The basic idea is that fusion represents a decrease in entropy, whereas splitting represents an increase in entropy.

Part I [26] makes brief mention of the notion of the temporal support $\vec{t}$ of the observer $\mathcal{O}_{\vec{\alpha}}$. This notion is developed to a greater extent in Part III [28]. For the reader who is familiar with the considerations of Part III, it is attractive to speculate that, given any arbitrary location in $\mathcal{O}$ -state space (that is, given any arbitrary $Q_{r,T}$, prior to our choice of temporal support $\vec{t}$), the most “natural” choice of temporal support $\vec{t}$ will be the one that maximizes the “splitting” and/or minimizes the “fusing” that will occur in the resulting tree diagram. A further attractive speculation would be that, given any arbitrary choice of temporal support, the time-dependent evolution of the object $\mathcal{O}_{\vec{\alpha}}$ forward in time will have the tendency to increase the ratio of splitting to fusing. In other words, we might speculate that the distal branches of a tree diagram that result in a “more natural” temporal support will tend to be more probable than distal branches that result in a “less natural” temporal support. In this manner, it may be possible to *derive* the second law of thermodynamics.

C. Relationship to Einstein’s ensemble interpretation

Einstein’s so-called ensemble interpretation – also referred to as his “statistical” interpretation – is discussed by Arthur Fine [13]. Little is known about Einstein’s concept of an ensemble as he did not develop his notions to any great extent. As Fine writes: “*nowhere at all* does Einstein say in any detail just exactly what this ensemble interpretation amounts to.” But in Einstein’s words (also taken from Fine [13]), the ensemble interpretation is summarized as the notion that “the

Ψ -function is to be understood as the description not of a single system but of an ensemble of systems” [13].

In the RHF, the Ψ -function is indeed to be understood as the description not of a single system, but of an ensemble of systems. In the RHF, the word “system” is replaced with a four-manifold $\mathcal{W}_{\vec{\gamma}}$; and the wavefunction of the Universe is represented, indeed, as an ensemble $E_{\alpha}^{\mathcal{W}}$ of these systems. The RHF, therefore, is an explicit demonstration of the feasibility of Einstein’s ensemble interpretation. Indeed, this is the inspiration for attaching the fancy word “ensemble” to $E_{\alpha}^{\mathcal{W}}$, instead of just calling it a “set” (which is, in fact, all that it is).

Why did Einstein not pursue his ensemble interpretation? What are the reasons for what Fine calls Einstein’s “cryptic brevity” regarding “the cornerstone of his own [Einstein’s] understanding of the quantum theory”? Is this one area – perhaps the only area – where Einstein lacked the courage of his convictions to pursue his intuitions down to their logical conclusions? I will not speculate here. But I will say that after nearly single-handedly achieving, say, two and a half revolutions in the field of physics – Brownian motion (one), GR (another), and his contributions to QM (give him half credit) – it would be a lot to expect him to have achieved yet another revolution, that is, the derivation of QM from GR.

D. Relationship to Cramer’s Transactional Interpretation

Cramer’s Transactional Interpretation of Quantum Mechanics. “The basic element of TI is the transaction describing a quantum event as an exchange of advanced and retarded waves ...” [7]. Relationship of the advanced and retarded waves to the forward and reverse paths φ^f and φ^r of the RHF? (in progress)

E. Relationship to the Feynman path integral formulation

In Part I [26], we show that the formalism of the RHF is equivalent to the formalism of the Feynman path integral formulation (FPI) (summarized in the Appendix to Part I; also see [8] [11] [12]). Let us now examine one of the “philosophical” similarities between the FPI and the RHF.

In the FPI, the wavefunction of a particle can be visualized as a collection of individual paths, each of which leads from one position $a(x, t)$ in spacetime to another position $b(x, t)$. Along any *individual* path, the particle is understood to have a *definite* position and momentum at each point along its trajectory. Philosophically, this is in violation of the Heisenberg Uncertainty Principle (HUP), according to which a particle cannot have a definite position and momentum at any given point in time. The reason that the FPI is not *explicitly* in violation of the HUP is that the wavefunction is never represented as a single path; rather, it is always a *collection* of many paths.

A similar situation exists in the RHF. Indeed, the RHF takes this “philosophical” violation of the HUP to the extreme. In the RHF, the wavefunction of the Universe is represented by a collection $E_{\alpha}^{\mathcal{W}}$ of manifolds $\mathcal{W}_{\vec{\gamma}}$. Within any individual manifold $\mathcal{W}_{\vec{\gamma}}$, not just one particle, but *every* particle can be under-

stood to have an *exact position and momentum* at *every* point along its trajectory. As is the case in the FPI, this represents a “philosophical” violation of the HUP. However, as with the FPI, the RHF does not explicitly violate the HUP for the simple reason that $E_{\alpha}^{\mathcal{W}}$ is never populated by just a single $\mathcal{W}_{\vec{\gamma}}$; rather, it is always a *collection* of many $\mathcal{W}_{\vec{\gamma}}$ ’s.

The de Broglie-Bohm (dBB) theory is similar to the FPI in the sense that it, too, works with individual particle paths. Therefore, the same general comments made here about the FPI can also be made about dBB.

VII. DISCUSSION

In this paper, we have presented the argument that the RHF offers a constellation of interpretational advantages (as listed in the Introduction) that are not to be found in any other single formulation of quantum mechanics. Certainly, many of these advantages are found individually within some of these other formulations. The consistent histories formulation, for example, is motivated primarily by the desire to devise a formulation that avoids any fundamental split between the observer and the system [20]. Everett’s relative state formulation, likewise, succeeds in this attempt. Both of these formulations, on the other hand, adopt quantum statistics as a fundamental assumption. We do not, of course, argue that it is *incorrect* to adopt quantum statistics as an assumption. Rather, we argue that it is advantageous, if only for the sake of preserving our sanity, to be able to think of probability in purely classical terms – as we do in the RHF – and then to see quantum statistics as a sort of “emergent phenomenon” that arises as a result of a change in perspective from one space ($\mathcal{O}$ -state space, in which probability is classical) to another space ($\mathcal{W}_{approx}$, in which evolution of an object under observation is governed by quantum statistics).

Perhaps the greatest interpretational advantage of the RHF is its *interpretational versatility*. We have seen how it is that the RHF lends itself naturally to *two* interpretations, one of which is motivated by positivism, the other by realism. The reader, therefore, is free to adopt whichever one of these interpretations that he or she chooses – but is also encouraged to contemplate the ease with which one is able to shift into the *other* viewpoint. These two interpretations also allow us to understand the EPR argument that quantum mechanics cannot be both complete and local. Furthermore, we have seen how the RHF allows us to conceptualize the difference between Bell nonlocality and Einstein locality, simply by focusing our attention on different mathematical entities within the RHF. In addition, we have also seen how several of the well-established interpretations and formulations – the consistent histories formulation, Everett’s relative state formulation (the MWI), Einstein’s ensemble interpretation, the Feynman path integral approach, and de Broglie-Bohm – each have their own “philosophical” connections to the RHF. We anticipate that other formulations, such as the TIQM, may also be easily translated into the language of the RHF.

It is only fair to examine the interpretational weaknesses of the RHF, so that we may weigh the price that we pay for its

advantages. Of course, “interpretational weakness” is a matter of taste. But let us suggest that the primary interpretational disadvantage of the RHF is that we must accept the notion that spacetime, *on the small scale*, is multiply connected and acausal. It will be a topic for Part III [28] (in progress) to argue that the general framework of the RHF simply would not work as a theory of quantum mechanics without this feature.

Perhaps one further psychological barrier to the acceptance of the RHF is the fact that the RHF requires the mathematical representation of the observer, that is, the three dimensional manifold $\mathcal{O}_o = \mathcal{O}_{\bar{\alpha}}$, to provide sufficient constraints upon $E_{\bar{\alpha}}^{\mathcal{W}}$ (that is, in its role as the boundary condition to $E_{\bar{\alpha}}^{\mathcal{W}}$) such that $E_{\bar{\alpha}}^{\mathcal{W}}$ can truly play the role of “wavefunction of the universe.” In other words, “everything that there is to know” about the world “out there” (external to the observer) is of necessity encoded, somehow, on the *internal structure of the observer itself*. It is perhaps difficult to imagine how so much information could be packed into so small a space! How might we answer such an objection? We would note, first of all, that Everett’s relative state formulation is likewise founded explicitly upon this concept. A rejection of the RHF on these grounds, therefore, also necessitates a rejection of the MWI, on the same grounds. Perhaps our best answer might be to suggest that this simply represents the limitations of the imagination. At this point, our imagination is the only tool available to tell us “how much information” must be encoded in $\mathcal{O}_{\bar{\alpha}}$ for it to play this role. One can easily argue that the “amount of information” required to specify the wavefunction of the Universe is infinite. One then realizes, however, that the amount of information encoded in the mathematical representation of $\mathcal{O}_{\bar{\alpha}}$ is likewise infinite [1]. It might seem that these two infinities should differ by many orders – but how many? This is where we see that this line of reasoning is difficult to develop with any mathematical rigor. Unless and

until this argument is presented with some rigor – quantifying the relevant “infinities,” for example – it would seem unfair to object to the RHF in its entirety solely on these grounds.

Although we have not discussed the particular features of the 4-geon model presented in Part I [26], or of the demonstration of the equivalence of the RHF and the FPI in that paper, a final objection that we will discuss here is that the details of the 4-geon model seem quite arbitrary. Our answer to this objection takes the form of an apology: the general strategy that we have adopted so far in devising the RHF is, essentially, one of “reverse engineering.” In other words, we started at the outset with the intention that quantum statistics should arise from our 4-geon model. It is our hope that the postulated characteristics of this model may be seen to arise from a deeper set of postulates governing the various mathematical entities of the RHF. Indeed, as we argue in Part III [28], this endeavor seems to be suited most naturally to the realm of Morse theory and other mathematical tools that form the basis of string theory. If this turns out to be the case, it may be possible to formulate the mathematics of string theory on the basis of the physical postulates of the RHF. (These physical postulates are summarized in the Discussion to Part I [26].) This can only be a good thing, considering that string theories have heretofore been “interpretationally challenged” by the lack of a grounding in any firm set of physical postulates.

Acknowledgments

We wish to acknowledge Ed Huff, Mark Hadley, zipperz, his evil twin alacrity, Tom Gutierrez, Scott C. Johnson, David Charles Hobby, vznuri, scerir, Travis Norsen, Eric Dennis, and others for their extremely helpful discussions.

-
- [1] How much information goes into the specification of a three dimensional object manifold, $\mathcal{O}_{\bar{\alpha}}$? Any manifold, of course, is a collection of an infinite number of points. Each point (x_1, x_2, x_3, x_4) is characterized by a metric, $g_{ab}(x_1, x_2, x_3, x_4)$, which is made up of ten independent components. Each of these ten components is a real number. And the number of bits that are required to specify just one real number, of course, is infinite.
 - [2] Bell, J.S. On the Einstein-Podolsky-Rosen paradox. *Physics* 1, 195-200, 1964.
 - [3] Bell, J.S. Speakable and Unspeakable in Quantum Mechanics.
 - [4] Bohm, David. A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables. I. *Physical Review*. Vol 85, No 2. (Jan 15, 1952) 166-179.
 - [5] Bohm, David. A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables. II. *Physical Review*. Vol 85, No 2. (Jan 15, 1952) 180-193.
 - [6] Bohm, David. The paradox of Einstein, Rosen, and Podolsky. *Quantum Th*, 611-623, 1951.
 - [7] Cramer, John G. *Reviews of Modern Physics* 58, 647-688, July (1986). http://mist.npl.washington.edu/npl/int_rep/tiqm/TI_toc.html
 - [8] Egli, Christian. *Feynman Path Integrals in Quantum Mechanics*. <http://www.hedweb.com/manworld.htm> <http://courses.physics.kth.se/5A1381/reports/egli.pdf>
 - [9] Einstein, A and Rosen, N. The particle problem in general relativity. *Physical Review*, 48:73-77, 1935.
 - [10] Everett, Hugh. “Relative State” Formulation of Quantum Mechanics. *Reviews of Modern Physics*. Vol 29, No 3. (July 1957) 454-462.
 - [11] Feynman, RP, Leighton, RB, and Sands, ML. **The Feynman Lectures on Physics**. Addison-Wesley Publishing Co., Inc., Reading, MA. 1963.
 - [12] Feynman, RP and Hibbs, AR. **Quantum Mechanics and Path Integrals**, (Boston: McGraw-Hill, 1965).
 - [13] Fine, A. **The Shaky Game: Einstein Realism and the Quantum Theory**. Second Ed. University of Chicago Press, Chicago, IL 1996.
 - [14] Hadley, Mark J. A Gravitational Theory of Quantum Mechanics. (Doctoral thesis) <http://www.warwick.ac.uk/~phsem/>
 - [15] Hadley, Mark J. *The Logic of General Relativity Derived from Classical General Relativity*. *Foundations of Physics Letters* Vol

- 10, No.1, (1997) 43-60.
- [16] Hadley, Mark J. Topology change and context dependence. *International Journal of Theoretical Physics* Vol. 38 (1999) 1481-1492
 - [17] Diemer, T. and Hadley, Mark J. Charge and the Topology of Spacetime. *Class. Quantum Grav.* 16 No 11 (November 1999) 3567-3577
 - [18] Hadley, Mark J. Spin half in classical general relativity. *Class. Quantum Grav.* 17 No 20 (21 October 2000) 4187-4194
 - [19] Hadley, Mark J. The orientability of spacetime. *Class. Quantum Grav.* 19 (August 2002) 4565-4571
 - [20] Isham, C. J. Quantum Logic and Decohering Histories. [arXiv:quant-ph/9506028](http://arxiv.org/abs/quant-ph/9506028) v1 16 Jun 95.
 - [21] Isham, CJ and Linden, N. Quantum temporal logic and decoherence functionals in the histories approach to generalized quantum theory. *J. Math. Phys.* 35 (10), October 1994 5452-5476.
 - [22] The online Everett FAQ. <http://www.hedweb.com/manworld.htm>
 - [23] C W Misner and J A Wheeler. Classical physics as geometry. *Annals of Physics*, 2:525, 1957.
 - [24] Valerio Scarani, Wolfgang Tittel, Hugo Zbinden, Nicolas Gisin The speed of quantum information and the preferred frame: analysis of experimental data. *Phys. Lett. A* 276 2000 1-7.
 - [25] Styer, Daniel F. et al. Nine formulations of quantum mechanics *Am. J. Phys.* 70 (3), March 2002. 288-297.
 - [26] Strayhorn, WD. The Relative Histories Formulation of Quantum Mechanics I: Introduction.
 - [27] Strayhorn, WD. The Relative Histories Formulation of Quantum Mechanics II: Interpretational Issues.
 - [28] Strayhorn, WD. The Relative Histories Formulation of Quantum Mechanics III: Further Mathematical Development. (Work in progress)